

INSTALLATION & MAINTENANCE MANUAL

Gautam Solar — All Module Series | Edition 2024

1. Safety Precautions

- Always wear insulated gloves and safety glasses during installation.
- Do NOT install panels in wet or stormy weather conditions.
- Modules generate DC voltage when exposed to light — treat as live at all times.
- Ensure all wiring complies with local electrical codes and standards.
- Use only certified MC4 connectors. Never use damaged connectors.
- Disconnect all strings before performing any maintenance work.
- Minimum 2 workers required for lifting panels above 500 Wp.

2. Pre-Installation Checklist

- Verify roof/ground structure load-bearing capacity (min. 25 kg/m2 for rooftop).
- Confirm roof tilt angle (ideal: 10°–30° for most Indian locations).
- Inspect all modules for shipping damage before installation.
- Check all junction boxes, cables, and connectors are intact.
- Verify module serial numbers match invoice and warranty cards.
- Ensure earthing/grounding system is in place before electrical connection.
- Confirm shading analysis — no shading between 9 AM and 3 PM.

3. Mounting Guidelines

Modules must be mounted using the pre-drilled holes on the aluminium frame or using approved mid/end clamps. The mounting structure must be galvanized or anodized to prevent corrosion. Minimum gap of 100mm between module bottom edge and roof surface is required for airflow and thermal management.

Indian Region	Latitude (approx.)	Optimal Tilt Angle	Expected Yield Gain
North India (Delhi, Punjab, UP)	28°–32°N	28°–32°	5–8% vs flat
West India (Rajasthan, Gujarat)	22°–28°N	22°–28°	6–9% vs flat
South India (TN, Karnataka, AP)	10°–18°N	10°–18°	4–7% vs flat
Central India (MP, Maharashtra)	18°–24°N	18°–24°	5–8% vs flat
East India (Bengal, Odisha)	20°–26°N	20°–26°	4–7% vs flat

4. Electrical Wiring

- Use UV-resistant, double-insulated DC cables (recommended: 4mm2 or 6mm2 for string lengths above 50m).
- Maximum string voltage must not exceed inverter's maximum input voltage.
- Install DC isolator/disconnect switch within 1m of inverter.
- All cable runs must be secured with UV-resistant cable ties every 300mm.
- Use ferrules for all cable terminations inside combiner/junction boxes.
- Label all strings clearly with string number and polarity.

5. Maintenance Schedule

Frequency	Task	Method
Weekly	Visual inspection	Check for soiling, bird droppings, damage
Monthly	Panel cleaning	Soft brush + clean water, early morning
Quarterly	Electrical check	Check Voc, Isc, IR scan if available
Bi-annual	Mechanical inspection	Check clamps, bolts, earthing, cables
Annual	Full system audit	IV curve tracing, EL imaging, inverter service
After storm	Damage inspection	Check for microcrack, broken glass, loose clamps

6. Troubleshooting Guide

Issue — Low Power Output:

Check shading, soiling on panels. Test Voc and Isc per string. Inspect for hot spots using thermal camera.

Issue — String Not Producing:

Check DC isolator, fuse, and MC4 connector continuity. Verify polarity (reversed polarity causes 0 output).

Issue — Inverter Fault:

Note error code. Common: grid fault, DC overvoltage, insulation fault. Contact inverter OEM support.

Issue — Hot Spot Detected:

Isolate affected module. Inspect with EL camera. Raise warranty claim if manufacturing defect found.

Issue — Water Ingress in JB:

Replace junction box. Check IP rating. Ensure cable glands are properly tightened.